Xander Byrne

Astrophysics PhD Student

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Research Interests

• Exoplanet dynamics • Rocky planet composition • Data science & machine learning

Education

Ph.D. in Astronomy, Institute of Astronomy, University of Cambridge 2023—
Dissertation: *Clues to exoplanet composition from stellar spectra*.
Supervisor: Dr Amy Bonsor

M.Sci in Natural Sciences (Astrophysics), University of Cambridge 2019–23
Graduated 1st class with distinction.

Publications

- **X. Byrne** & A. Bonsor (2026).
  The true masses of radial-velocity exoplanets constrained by stability.
  RASTI, 5. doi:10.1093/rasti/rzag065.
- **X. Byrne**, C. M. Guimond, A. Bonsor, et al. (2026).
  The Barnard's Star Planetary System: stability, composition, and evolution of four sub-Earth exoplanets.
  MNRAS, 550(2). doi:10.1093/mnras/stag1207.
- **X. Byrne**, A. Bonsor, L. K. Rogers et al. (2025).
  Finding rare classes in large datasets: the case of polluted white dwarfs from Gaia XP spectra.
  RASTI, 4. doi:10.1093/rasti/rzaf044.
- **X. Byrne**, A. Bonsor, L. K. Rogers et al. (2024).
  Semi-supervised spectral classification of DESI white dwarfs by dimensionality reduction.
  MNRAS, 535(3), 2246-2259. doi:10.1093/mnras/stae2478.
- **X. Byrne**, R. A. Meyer, E. P. Farina, et al. (2024).
  Quasar Island - three new $z \sim 6$ quasars, including a lensed candidate, identified with contrastive learning.
  MNRAS, 530(1), 870-880. doi:10.1093/mnras/stae902.
- **X. Byrne**, O. Shorttle, S. Jordan et al. (2024).
  Atmospheres as a window to rocky exoplanet surfaces.
  MNRAS, 527(4), 10748-10759. doi:10.1093/mnras/stad3914.
- A. Mata-Sanchez, E. P. Farina, A. J. Gloudemans, et al. inc. **X. Byrne** (2026).
  To be lensed or not to be lensed: on the nature of the alleged high-z lensed quasars J0109−5424 and P170+20.
  ApJ, 1010, 15. doi:10.3847/1538-4357/ae9d7e.

Talks

Invited Talks:

- **White Dwarf Meeting – University of Warwick** Feb 2026
  Big-Data Methods for Polluted WDs – Past, Present and Future
- **ExoCoffee – MPIA, Heidelberg** Feb 2024
  Atmospheres as a Window to Rocky Planet Surfaces
- **Data Intensive Science Seminar Series – Maxwell Centre, Cambridge** Feb 2024
  Contrastive Learning and High-Redshift Quasars

Contributed Talks:

- **EuroWD – ISTA, Vienna** Aug 2026
  Big-data methods for polluted white dwarfs: past, present, and future
- **Origins 2026 – Paris** Jul 2026
  The Barnard's Star Planetary System: interior structure and water capacity of four sub-Earth planets
- **UK White Dwarf Science Meeting – Queens' College, Cambridge** Mar 2026
  Calcium Needles in Hydrogen Haystacks – Identifying polluted white dwarfs with Gaia
- **Wednesday Seminar Series – Institute of Astronomy, Cambridge** Oct 2024
  Spectral Classification of White Dwarfs by Dimensionality Reduction
- **Astro Data Science Discussion Group – KICC, Cambridge** Jan 2024
  Contrastive Learning and High-Redshift Quasars
- **Galaxy Coffee – MPIA, Heidelberg** Sep 2022
  Searching for Gravitationally-Lensed High-Redshift Quasars with Unsupervised Machine Learning

Awarded Telescope Time

As Co-I:

- *White dwarfs as a means to probe core formation in exoplanetary systems*; PI Rogers
  65h VLT/X-Shooter spectroscopy (115.28F4)
- *Characterising a $z \sim 6$ gravitationally-lensed BAL quasar candidate*; PI Meyer
  2.9h VLT/X-Shooter spectroscopy (113.26CY)
- *Measuring the mass of the SMBHs powering two lensed $z \sim 6$ quasars*; PI Meyer
  4.23h Gemini (DDT) FLAMINGOS-2 spectroscopy (DT-2022B-029)

Professional Service

Seminars:

- **Organiser, Exoplanet Seminar Series** Oct 2026—
  Inviting speakers and coordinating talks for an interdepartmental seminar series on exoplanet science.
- **Organiser, Wednesday Seminar Series** 2024–26
  Coordinating and hosting this long-running seminar series, given by PhD students and postdocs at the Institute of Astronomy.

Conferences:

- **LOC, Debris Disk Connections Workshop** Jul 2026
  Assisting speakers and chairs at a large conference with presentations, mics etc.
- **Organiser, UK White Dwarf Science Meeting** Mar 2026
  Coordinating invitations, payments, scheduling, and talks for a national one-day meeting.
- **LOC, *IoA50* conference** Jul 2024
  Assisting speakers and chairs at a large conference with presentations, mics etc.

Other:

- **Organiser, PhD/Postdoc Jamboree** 2024—
  A social 'introduction' event for the Institute's PhDs and postdocs.
- **Co-chair, LGBTQ+ Focus Group, Institute of Astronomy** 2025–26
  Organising departmental events.
- **Organiser, Undergraduate Journal Club** 2023–24
  A journal club for undergraduates, also offering career advice.

Grants, Awards & Prizes

- **IoA Community Service Award** Institute of Astronomy
  For 'exemplary contributions to Public Outreach and the Wednesday Seminar Committee'.
- **MASt/Part III Bursary** (£3500) Institute of Astronomy
  Funding the conversion of my Masters' thesis to a publication over the course of a summer internship.

- **MPIA Summer Internship** (€3750) Max-Planck-Institut für Astronomie
  Funding internship: *Searching for Lensed High-Redshift Quasars using Unsupervised Machine Learning*.
- **Winifred Georgia Holgate-Pollard Memorial Prize** (£1650) University of Cambridge
  Awarded for the best performance in the Natural Sciences (Astrophysics) examinations.
- **John Spencer Wilson Prize** (£500) St Catharine's College, Cambridge
  Awarded in recognition of exceptional performance in the Natural Sciences examinations.

Teaching

Supervisor, Part II Astrophysics, University of Cambridge 2023—
- Conducted small-group teaching, marking solutions
  - *Relativity* (GR): 18 students
  - *Stellar Dynamics & Structure of Galaxies* (orbits & potentials): 8 students
- Produced popular revision notes (~ 2500 hits per year)

Miscellaneous Experience

Technical Experience:
- **Programming Languages:** Python, C++, MATLAB
- **Astronomy Software:** EsoRex, EsoReflex, DS9, TOPCAT, PyTorch, TensorFlow
- **Development:** Primary developer and maintainer of the open-source `cinemas` package
- **Datasets:** Gaia, DESI, SDSS, GALAH

Industry Experience:
- **Internship at Featurespace** Feb–Aug 2025
  6-month placement in the R&D division of a tech company. Implemented and tested state-of-the-art innovations in AI research into the company's extensive codebase.

Outreach

- **Cambridge University Astronomical Society** Oct 2026
  Talk on the Barnard's Star planetary system. This society has previously hosted talks by Hoyle, Eddington, Bell Burnell, and Queloz.
- **Peterborough Astronomical Society** Oct 2026
  An outreach talk at the PAS.
- **Pint of Science** May 2026
  A public talk in The Alexandra Arms pub, Cambridge: *The 10 nearest stars to the Sun*
- **Asdan Summer School, St Catharine's College, Cambridge** Aug 2025
  Outreach talk: *Worlds Anew: Exoplanetary Astronomy*
- **IoA+KICC Open Day** 2024—
  Organising and running activities, talks, and demonstrations for the general public.
- **Cambridge Hands-On Science (CHaOS) event: *Crash Bang Squelch*** Mar 2024
  Outreach talk: *How to find an Alien Planet*
- **Birmingham Schools Science Network** Sep 2023
  Outreach talk: *Worlds Anew: Quasars, Exoplanets, and the Search for Life*.
- ***The Naked Scientists*, BBC Radio 5 Live** Mar 2023—
  National radio show and podcast. Numerous appearances both live and in pre-recorded segments
- **Public Open Evenings, Institute of Astronomy** 2022—
  Visits for the general public, involving a public lecture (which I gave in 2025 and 2026) and stargazing.
- **HiSPARC Project** 2018–22
  Teaching sessions and sparking research projects for high-school physics students, particularly girls
- ***Varsity Sci 2021*** Oct 2021
  Outreach talk: *The Messier Catalogue: 110 objects in 10 minutes*.
- **Cambridge University Physics Society** Aug 2020
  Outreach talk: *Negative mass: matter that's lighter than light*.

Press Coverage

- **University of Cambridge**: *Astronomers find nearby planets to be small, strange, and utterly uninhabitable* Jun 2026
- **NOIRLab**: *AI Aids in Search for Cosmic Gems in Ocean of Data* Jul 2024